

UGUIDU

P R E S E N T A T I O N

Presented by

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UGUIDU

AI-POWERED INCLUSIVE MUSEUM VOICE ASSISTANT

UGUIDU is a multimodal AI voice assistant. Enabling visitors to interact with artworks through natural speech. The system combines voice recognition, computer vision, and retrieval-augmented generation to deliver real, contextual conversations about museum artifacts in Dutch and English.

Built and engineered over 20 weeks as part of an AI Engineering internship at BUas Experience Lab.

PROBLEMS

PROBLEMS

1

Visitors Disengage

Labels give 3 lines of text. Artifacts have centuries of story. There is no bridge between them.

2

Language Barriers

Dutch artifacts, international visitors. One static experience that fits nobody well.

3

Knowledge Stays Locked

Museums hold rich content. Most of it never reaches the person standing right there.

CLIOZOOFT

Solution 1

Point the camera. AI identifies the artifact instantly from 31 learned classes using visual AI.

Solution 2

Ask anything in Dutch or English. No buttons, no menus. Just talk.

Solution 3

AI retrieves museum knowledge and responds in real, contextual conversation.

SOLUTIONS

WHAT WE BUILT



UGUIDU runs as a web-based interface, live, deployed, and tested on BUas servers. Visitors scan an artifact, speak naturally, and get a real conversation back.

A second version is in the pipeline: a hardware-first experience built around the Rabbit R1 device, designed for hands-free, always-on museum engagement. That is the next chapter.

BIRTH OF UGUIDU

1

MARCH
2026

European students and lecturers from multiple countries tested the system. 17 participants total from the Erasmus BIP program.

2

APRIL
2026

Expert testing with accessibility specialists and users with disabilities. Focus on usability and inclusivity of the voice interface.

3

JUNE
2026

Live museum test with 15+ real users: senior visitors and Stedelijk Museum Breda staff. Longest conversations yet, zero errors.

VIEWPOINT

› Built full RAG pipeline:
BGE-M3 embeddings +
pgvector + Qwen3-14B —
from scratch

› Invented See-Think-
Wonder museum
educator prompt — same
model, transformative
results

› Live deployment on BUas
server — a running system,
not a prototype

› Debugged CLIP
classification after
museum playtest exposed
real-world edge cases

THE MOMENT IT BECAME REAL

Late May 2026 — UGUIDU went live in Stedelijk Museum Breda.
Real visitors. Real conversations. No instructions needed.

APPROACH



1

APPROACH 1

Visitors just pointed and talked.
Zero instructions needed.



2

APPROACH 2

Dutch and English mid-conversation.
Zero friction, fully automatic.



3

APPROACH 3

Camera angle problems surfaced.
Debugged and fixed live.

THE VERDICT

“

Everyone had longer conversations, used images, sound and text — and there were no errors.

Words used by users: 'PROMISING' and 'POTENTIAL'.

My personal feeling was that we made a genuine step forward.

Adriaan van Liempt — Internship Supervisor, BUas Experience Lab

PROMISING

POTENTIAL

STEP FORWARD

WHAT I LEARNED

1

REAL ENVIRONMENTS BREAK EVERYTHING

Camera angles in the museum were nothing like our test setup. That gap is where real engineering starts.

2

PROMPTING IS AN ENGINEERING DISCIPLINE

The See-Think-Wonder educator prompt transformed AI response quality. Same model. Better framing.

3

DESIGN FOR PEOPLE WHO DON'T READ MANUALS

Every friction point removed was a design decision. The best interface is no interface at all.

WHAT'S NEXT

1

Knowledge Base

Expand and enrich museum content. More artifacts, deeper descriptions, and stronger links between related objects so conversations go further.

2

VISION MODEL

Train a dedicated vision model on real museum conditions. Calibrated at visitor distance, not 10cm. More classes, more accuracy.

3

CONTROLLED TESTING

Structured testing environment with defined metrics. Automated integration tests, reproducible playtest conditions, and formal inclusivity measurement across all nine dimensions.

THANKS

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